

Assignment 01

01 Write down the negation of each of the following statements.

- (i) Product of two odd numbers is odd.
- (ii) If mn is odd then m is odd and n is odd
- (iii) If m is odd or n is odd then mn is odd
- (iv) If mn is even then m is even or n is even
- (v) Either m or n is even.

02 Write down the contrapositive of each of the following statements

- (i) If mn is odd then m is odd and n is odd
- (ii) If m is odd or n is odd then mn is odd
- (iii) If mn is even then m is even or n is even

03. Prove or disprove each of the following statements, where m, n are integers.

- (i) Product of two odd numbers is odd.
- (ii) If mn is odd then m is odd and n is odd
- (iii) If m is odd or n is odd then mn is odd
- (iv) If mn is even then m is even or n is even

04. Prove the following statements.

- (i) Sum of any two rational numbers is rational.
- (ii) Product of any two rational numbers is rational.

05 State whether each of following statements is true or false. Justify your answers.

- (i) Product of any two irrationals is irrational.

- (ii) Sum of any two irrationals is irrational.
- (iii) Sum of a rational number and an irrational number is irrational
05. Prove that between any two real numbers there is a rational number.
06. Prove that between any two real numbers there is an irrational number.
07. Let m be an odd positive integer. Prove that $\sqrt{2m}$ is irrational.
08. Find the base 10 representation of $(453)_7$
09. Find the base 2 representation of 214 {i.e. $(214)_{10}$ }.
10. Let $N = a_m 10^m + \cdots + a_2 10^2 + a_1 10 + a_0$, where $0 \leq a_i \leq 9$ for $i = 0, 1, \dots, m$ be the decimal expansion of a positive integer N .
- (i) Prove that $10^i - 4$ is divisible by 6, for every positive integer i .
- Deduce that N is divisible by 6 if and only if $a_0 + 4a_1 + 4a_2 + \cdots + 4a_m$ is divisible by 6.
- (ii) Prove that N is divisible by 8 if and only if $a_0 + 2a_1 + 4a_2$ is divisible by 8
- Find similar criteria for divisible by 16.
11. Let $N = a_m 10^m + \cdots + a_2 10^2 + a_1 10 + a_0$, where $0 \leq a_i \leq 9$ for $i = 0, 1, \dots, m$ be the decimal representation of positive integer .
- Let $T = a_0 + 10a_1 + a_2 + 10a_3 + \cdots +$.
- Show that $N - T$ is divisible by 99.
12. International Standard Book Number (ISBN) appeared in most books consists of nine digits $a_1, a_2, \dots, a_9$ followed by a tenth check digit a_{10} , which set as the remainder when $\sum_{k=1}^9 k a_k$ is divided by 11. Note: If the remainder is 10 then it will be written as X.
- The ISBN number of the book “Elementary Number Theory, David M Burton, Sixth Edition” is ISBN 0-07-061607-8. Check whether the check digit is correct.
13. Find the minimal period of the digital representation of the rational number $\frac{1}{17}$.

14. Find the relevant rational numbers for the following decimal representations

(i) $23.456\overline{789}$

(ii) $0.\overline{101}$

(iii) $0.\overline{00001}$

15. Consider the sequence $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$. In how many terms of this sequence have the terminating decimal expansions from term 1 to term 51.

16. How many zeros are there at the end of the decimal expansion of $51!$? (Factorial 51)